

Limited DVs and GLMs

Estimating a Probit or Logit Model

You will be working with the dataset *urbanbias*, which is from the paper:

Cameron Ballard-Rosa (2016) "Hungry for Change: Urban Bias and Autocratic Sovereign Debt Default." *International Organization*, 70(02), 313-346.

This article focuses on urban unrest and increasing food prices in non-democracies, and how this increases the probability of sovereign default (i.e. government does not pay their debts). Default (*default_lead*) is a binary outcome variable, which is coded as one when a government is in default, and coded as zero otherwise. Food imports (*foodimp_gdp*) and urbanization (*wdi_urban*) are the explanatory variables of interest. Note that the analysis is only required for a subset of observations, non-democracies (i.e. *democ*=0). The original paper uses (too) many control variables. The following are reasonable ones to include in a model: log GDP per capita (*ln_gdp*), growth (*chg_gdp*), debt-to-GDP ratio (*debtgdp*), inflation (*inflation*), and trade dependence (*trade*).

1. Estimate separate logit and probit GLM models with default lead as the y, and including only food imports as an explanatory variable.
2. Provide a full interpretation of the results from the above models. What can you infer from the output of the regression? Provide substantive interpretations in the form of marginal effects, predicted probabilities, and odds ratio (for the logit model). Construct at least one visual representation of these effects.
3. Repeat step 1, but now include urbanization into the model. Provide a full interpretation of urbanization's effect.
4. Now include all control variables mentioned above into the model. Which model is better: the model with only urbanization and food imports or the model that also includes the controls?
5. Finally, interact urbanization and food imports and include that interaction into a logit model. Is there a conditional effect? Construct a visual representation of this potential effect.